

Yes. So, I wanted to first do a summary, and then the next steps for the project.

So, so far, like I sent you in the graphs before, that from the frustration 0 to 0.5, all of the folding was 2-state, but then when the frustration was 0.6, it became 3-state folding.

And after 0.6, and when F became 0.7, it was a kinetic trap at the second transition state, because the intermediate was very low in terms of free energy.

And when I... in the folding barrier height versus the frustration factor graph.

There was an optimum level of frustration at 0.65, but I have one question in terms of like, analyzing the simulation, and that is, how can I understand the contact map graph?

Like, when I generate it, then what can I learn from the contact maps?

So basically, you calculate the contact map, and the first thing you have to know which state you are calculating. Is it a transition state or intermediate state?

I did the transition state contact maps. Transition state, but you also mentioned that when you, you have this fraction, frustration factor, larger than 0.6, you are going to observe, an intermediate state, right? Yes.

And, at that time, I would expect that you would have two transition states, because, you have two,

energy barrier. So, at that situation, are you still calculating the transition state or the intermediate state?

I made the contact maps for both transition states.

The first one. Both transitions? Yeah. Okay. And

Okay, so, let me, let me go back to your result. When you, increase the frustration factor from 0 to 0.6, you, you didn't observe the intermediate state, right? Yeah, 0.6 was the first time you see the intermediate.

Okay, and

And, you, see the folding barrier decreasing as you, increase the frustration factor. Okay, good. So that actually means when you try to, add the non-native interaction into your system, those, non-native interactions can actually help the protein to fold, because you decrease, the folding barrier, right?

So this is very, I think it is a good result, to, to, see that, the, non-native interaction are not always bad, right? So it can actually help the protein to fold. But indeed, this...

finding or this statement, were actually observed by a previous study, but not quantitatively. I mean, because none... you... you observed that, the frustration increased to 0.6, and then there...

or there is an intermediate state. When the intermediate state, comes out, it can basically, slow down the protein folding, because you actually have a kinetic traps, along the folding pathway, right? But I, I would like, to know when,

I would assume that when you increase the frustration factor to, a very high value, and you basically couldn't observe the native structure of the, of this protein during the simulation. Have you ever... have you observed this phenomenon in your simulation? I increased the frustration factor to 1.

4, or 1.5.

And then I made the temperature very low, and the protein was...

still able to fold, but the temperature was so low, like, I think even in negative Kelvin, so it didn't make sense.

So I kept on... Negative... negative Kelvin. You mean temperature? No, no, I mean, like, when I had to change the MDP file to a lower temperature, it was very low, like, it was going to become negative, so I thought that didn't make sense to continue.

Okay, and currently, I'm actually reading the graph, the folding barrier head, versus the frustration factor, and I saw that you increased the, the, this frustration factor up to 1.1, right? Yeah.

And... and this folding barrier increased, actually. So I... Yeah, it started going higher.

What does this, free energy profile looks like?

Let me see if I have it here...

It... I don't know if, it's on my other computer, but the free energy barrier was very high, and it was very rugged, like, it was, it wasn't smooth at all, but it was still on the folded side, at low temperature.

So you can still observe the full data state, right?

Okay.

Let's see, okay. Okay, let me, let me go back to your, question. So, how could you understand this contact map, right? Yeah. Okay, firstly, this is a two-dimension, plot. It's actually described, whether or not this contact is formed at the...

In the transition state, right? And, the, red color indicate a high probability.

for this contact to form, and, and of course, the blue color indicates a low probability to fall, for this contact to form. Okay.

So you, you, and, and you probably, for the simulation part, you probably already know that the transition state is not a single state. It's actually, contains many, different structures. So basically, you cannot use one structure to describe the structural property of the transition state. So that's the reason you introduced the contact map, and those contact maps is actually, an average, description of the structure, or transition state structure.

So you, as you see that this is actually the, the probability, like, otherwise you would see that, one zero, like, contact is the form.

that's the 1, conduct is not formed, let's say that would be 0. But... but you see that the, the... this value is in between 0 and 1. So that actually means in some structure, it is formed, in some structure it is not formed, right? But for the single transition... for the... for the... for the transition state, right? So that actually means the transition state

as,

contains many, many different structures. And in transition state, the protein looks, different, from the native structure, and also different from the unfolded structure. Because ideally, when you, observe the protein in unfolded structure, you probably see a very low contact formation, a low fraction of contact formation in unfolded state. But in transition state, you would see that some contacts are likely to be formed, some contacts are not likely to... are unlikely to be formed.

So based on this, this kind of map, I think you should have a feeling of, which part, of this protein, is formed, which part, the protein is, is forming the native slack structure, in transition state. But why transition state is so important? You can think about this picture, that...

when you go over the transition state, the folding can be very fast, right? So it's like moving downhill.

So, the transition state is the bottom neck step.

a bottleneck state, for this protein folding. So protein has to spend a lot of time to go past... a very long time to go past the transition state. But after overcoming this translation state, the folding can be very easy.

Hello? So you can... so the structure... so that's the reason the structure is so important, because, people, are very interested in, why, in some cases, the protein falls so slowly, although it falls extremely fast in the cell, but,

It's not like, you know, folding... when you have the one-dimensional chain, and all of a sudden, the one-dimensional chain becomes a three-dimensional structure.

It also... it always takes time to, to fold for this protein. So people would like to understand why, the protein folds so slowly. So in order to understand why, we have to, look at the structure of the transition state. So at least you would know that, for example, if these two contacts

are forming a, you know, a very high probability in transition state. So that actually means that this contact may be, they...

bottom neck.

Stop The formation of this contact may be the bottleneck for this protein folding, because after

This contact is formed.

the system is actually at a transition state, and the system can go downhill to the final native structure. So this is the logic. So, in order to understand that, I think you, you, you, you, if you,

remember this paper, this, this, I sent to you the Journal of Molecular Biology, and they put the secondary structure, plot.

Just near to, the X and the Y axis. So that should give you an idea that, in transition state, which part of the protein

Has already been folded.

So that actually can give you a feeling. But, but,

Even, you look at this kind of map, it's showing in 2D, two-dimension. So this actually reflects the, information of a three-dimension.

Because you have a... you have a contact map. You... you have a... there's

three-dimensional contact. So, so, because you list all this contact map, but I would suggest you to do one more thing, that,

you don't add any frustration factor, you have that result, you know how the structure looks like in frustration state. And what you can do is, basically, you can,

You can, subtract your current

contact map with a certain degree of fraction... of a frustration factor to the, original contact map. Okay. And you will see how the, structure, changed.

For the transition state, as you add in the frustration factor.

So that should kind of give you an idea that how the native and non-native interaction changes the structure of a transition state when you add the frustration factor. So that's one thing. The second thing is, probably they will be a little bit difficult for you to do, but I would suggest you to do that. So now that all the contact maps are native contact.

You know the native contact, right? So the contact is actually forming in, the native structure.

Yeah. So, so you look at the contact map, it only reflects the...

native contact. Yeah, yeah. So it does not reflect non-native contact. So my suggestion is, you... you... you... the first thing I already suggest to you, the second thing is.

You can calculate the contact based on the distance.

For example, yeah, so for example, I don't care whether this residue I, amino acid I and amino acid J, whether they are in the native contact or not, I don't care. So, I just...

introduce a cut-off criterion. For example, I would suggest, like, 7.5 Å. So during the simulation,

I have a lot of simulation frame, right? So, I will observe this, distance between the residue I and the residue J. Whenever they are... this residue distance falls in, 7.5 Å.

I will... Assign, a contact For these two residue pairs.

Yes. And in this case, I don't have to worry about whether this I and J they are native contact or not, I just, you know, I just calculated the distance, based on... I just calculate, the, the, the con... the distance between the residue, I and the J. Whenever they are falling in the 7.5 Å.

I assume they are forming, a contact. And this contact does not have to be a native contact, it can be, non-native contact. Or, in the native... in the Go model, purely Go model, purely Go, and you can still have this non-native contact.

Okay. For the inner transition state.

Because although they are not stable, they are not having any stabilization, because in GoModel, you only have the attractive interaction between the residue I and J when they are in the native contact list, right? If this I and J are not in the native contact list.

Because these residue are forming... are moving randomly.

So, in some simulation frame, they accidentally become near to each other.

But you... but... but because there are no attractive interactions between this residue I and J, so when you average over the whole simulation frame, you would see the probability the contact probability for this residue energy are not high, because this is just accidentally contact with each other. But for the, model with frustration factor.

Yes. And, the INJ can still accidentally

Be contact with each other.

But they have another option, that this INJ are actually having attractive interaction between each other.

So they can actually form the... I wouldn't say very stable, but moderately stable interaction.

So they, they would have, they, they, they, they, they would have...

probably change the structure of the transition state. So this is the two analysts I would suggest you to do. One is based on the native contact. The other one is

Just based on the distance, and you have, a native contact map, you have a contact map.

Yeah. I'm not sure if you understand. No, I understand. So we add the, different one based on the distance cutoff of 7.5 Å.

And then I create a new contact... 7.5. Yeah.

7.5 or 8, I think, 7.5 should be fine. And because of the final... Final value, for, defining the contact, is not so important. What you are going to see is, a trend of how, the frustration factor changes the structural, transition state. So the exact value, of that criterion should not be very important. As long as we say that, it's reasonable, I mean, the result should be, convincing, okay? And this is the, this is the first thing. The next step, I would suggest you to do is, as I mentioned,

So this non-native interacting investigation has been partially done by previous, you know, researchers in this field. But one of the things they haven't done is

Maybe it's a little bit difficult for you to understand, but, I will, try to, make it as simple... make this statement as simple as possible. So...

When the protein is synthesized, in the cell,

they have the sequence, right? They have 20 different amino acids, you know, placed in the different position in the sequence.

And once they are synthesized, the nature knows
Whether or not this chain can be folded or unfolded.
Right? Because the sequence determines the foldability, foldability of this chain. Yes. Okay?
And, why you can do that, why can do that? That's simply because, the interaction.
the favorable interaction in the sequence, and they like to fold this one-dimensional chain in
a three-dimensional structure. So the sequence information is very important.
And, in the first, part of your project, you already demonstrated
that the non-native interaction can also be important for this protein, right? If we only
consider the native contact or native interaction in this model, we can still fold this protein.
But, we cannot afford this protein so efficiently, because when you add a non-native
interaction, you basically decrease the, the, the energy barrier for the protein to fold. And,
And when you add this non-native interaction.
you actually, consider the sequence information for this protein. So how did you do that? You
introduced the, MJ metric, right? Yep.
And this... the MGM metric is based on the experiment, and one simple rule is when you
have two hydrophobic residue.
And this interaction strength may be, should be stronger than a residue pair coming from
one hydrophobic list and one hydrophilic.
Federalic list, right? Okay, this is the syncy information. So you can think about this, situation
if
We still have this native structure, we still have it.
But we changed the sequence.
We changed the sequence. That...
How we... how we are going to change the sequence in this model? That we just randomly
change
the... I'm geometric.
That maybe in a situa... to change to a situation, maybe the hydrophobic pair will have a
weaker interaction than a hydrophilic pair.
Yeah. And, and then you can use that metric to... Simulate your goal model.
And to see if that non-native interaction can
Still, makes the protein fall so inefficiently.
I would, I would assume that that new sequence, It's not working as
good as, the original one. So would you randomly change the values in the MJ matrix, or
how would you choose what value to make it?
I, actually, I, my, my, my idea, firstly, is, so, so, I, I would, say that my, my, my, my idea is like
this.
We all know that the sequence, the native sequence, determines the protein to fold.
But I would also assume that the native syntheses also optimize the non-native interaction to
make sure the protein falls so efficient. So that's our,
hypothesis or assumption, okay? And we have already demonstrated that.
At the current situation.
the purely gold model can fold the protein very efficiently without considering the non-natal
interaction, and after adding
certain degree of non-native interaction. We can still fold the protein, and the folding can
become even better, because the non-native interaction accelerates the protein fold, protein
to fold.
Okay, so the next step, I would assume that, this non-native interaction... since the
non-native interaction can accelerate the protein to fold, what if I just change the non-native,

interaction by changing the, the, the, the, the MGM metric. Yeah, yeah.

So, so, let me, let me give you a running example. For example, in the native, in the translation state.

And, probably you will, observe, after, after doing this analysis for the contact and not non-native contact, perhaps in some, situation, you can observe a situ... You can observe, a phenomenon that in the transition state.

I'll just take it, for example, residue 2,

is a hydrophobic residue. Residue 20 is also a hydrophobic residue. They are not forming native contact in native structure, but in the transition state.

You probably can observe they are forming contact.

And these two are hydrophobic residue, both hydrophobic residue. They are formula contact because, in MDM matrix, you give them, a large

value. Yeah. So they are likely to, to form a contact in the... during the simulation. And so this one is very important to stabilize the transition state. So that actually means that, this one

Can be, good for the protein to fall, because you basically stabilize the transition state, and you will decrease the folding barrier.

So, if we change the MBGM metric.

For example, 100 is the residue 5, and 20 is going to have, a week. interaction.

And, in that situation, if you still introduce a same level of frustration.

And you are not going to observe the, the contact form in the translation state.

So that actually... so that actually can tell you,

The, the protein sequence is trying to optimize the...

The native contact and the native interaction and the non-native interaction at the same time.

So that's what I... so that's what I mean. But in practice, how... how could we, change the MGM metric,

Honestly speaking, in order to, exaggerate this, in fact, what we are going to do is we gotta, we just completely

reverse the trends of the MGM matrix. I mean, in the original MGM matrix, we have a hydrophobic residue energy interacting strongly with the hydrophobic residue.

But in order to exaggerate the infect, in the opposite way, that you are going to, say that this hydrophobic residue is going to,

interact weakly with a, hydrophobic residue. I, I think that would be, the, the, the optimal situation. But the problem is, we do not know how to.

Completely reverse the trend.

So we only know this, qualitatively, we know that that situation, you know, the hydrophobic residue, our pair are... are interacting, more strongly,

But... but we do not know, how we can, reverse this, this MGM matrix, so that's the reason I would propose that perhaps we just randomly, change the... change the MGM matrix.

Okay. Like, redistribute the value in the metric. Yeah, okay. That's what I'm currently thinking.

So I'm not sure if you... maybe you have a better, you know, suggestion regarding all this. Or maybe we can ask a chat GPT.

I'll do a favor.

Yeah. So, I also had a few next steps for the project, so... Okay.

So, one is that I want to visualize what the intermediate is like, and also I want to validate that the transition state, the second one, is actually... that the probability of folding there is 0.5, so it's a valid transition state.

So, what I want to do is commit an analysis, which is what I found online, which I can run short simulations at the transition state.

And calculate the probability of folding to see that

our free energy landscape is actually a representation of transition state. That's actually 0.5%. Okay, okay. So, so, so, I know that what you are going to do, so you take out the structure from a transition state, and then you run a very shallow simulation. For example, you have a 100 structure from a transition state. You would do this shallow simulation to see if finally this structure arrives

So, unfolded state or folded state, right? Yeah.

Yeah. Okay, and that's actually give you a quantitative, because it's called a, probability of commitment. Yeah. Right? Yeah, P4, right? Yeah, P4.

Yeah, I think, I think some of them have done this similar, analysis before, but I'm not sure if, they have done the similar thing to non-native interaction, but I would assume that, there would be, a good try.

Because the folding from the free energy landscape, you only have the transition state defined by the

I know that would be difficult for you, to be honest. Defined by some more dynamics.

Defined by the folding free energy. Yeah, the free energy. Yeah, so...

In the real system, in a real system, the free energy landscape should be very, very complicated, because,

I don't know how to explain, because the system have many, many atoms.

Right? And, in our situation, we have a metamatic beat.

And each... in order to

defund... in order to precisely determine or define a status of this beat, we need a three number. So that number is called a degree of freedom.

So you have 100... if you have a 100-bit, you are going to have a

You are going to introduce 300. Yeah. So for the free energy landscape, you are going to have a 300 degree of freedom.

But nowadays, the quantity will only have one dimension. We only use one number, which is fraction of native content, Q, right? To describe the free energy landscape.

So this is the huge... Simplification.

for this system.

And, so it does not guarantee that the transition state you observe from folding free energy is the real transition state. Yeah, so that's why I wanted to validate that it's real. Yeah, so I would assume that when you have the pure goal, and they probably will match very well.

But when you try to add the non-native interaction, maybe they... they start to, debate. I mean, the P4 cannot... should... probably will be lower than 0.5. Ideally, it should be 0.5, right?

I think that would be a good direction. But one thing I'm a little bit worried is, I'm not sure if someone has done this before. If someone have done it before,

We, there would be no, good

Chance for us to do it.

to repeat, actually, yeah? I think it should be,

You have done the, literature review, like, you search online and... Yeah, yeah, I searched online for it. Have you ever... I didn't see, I didn't see much of frustration included, but I'll continue searching to see if someone did this before.

Okay, okay, good. Yeah, I think this would be a very good, direction to go, and, shorter simulation,

But I wouldn't suggest you to do this short simulation for every value of a fraction factor. Like, you just pick all the...

Yeah, several, like, 0.1 0, 0.1, 0.2345, I think 5 is enough. And, and, and, meanwhile, you should have to, run, like, a 100 simulation for, each value of...

I, I, I think that would be...

A very good, direction. Yeah, yeah, good. Yeah.

So, the second thing I wanted to do is also see what the intermediate state is like, especially from 0.6, because that was the best result for the intermediate.

So, I want to see if it's a singular state, because sometimes the intermediate can have a collection of sub-states, not just one single state. So, I will use frames from the intermediate range, which from, I think, Q is 0.6 to 0.75.

And cluster, so use structural clustering to figure out how many... if it's a sub-state or a single state.

Okay, you mean you want to do the structure analysis for the intermediate state? Yeah, for the intermediate state at F is 0.6.

Yeah, that's, that's actually good. I think, when you, try to, analyze the contact map, so the second way of, doing the contact map, we talk about, the analysis for the transcription state, you can still apply the similar strategy to analyze, to analyze the intermediate state. I think from that, probably.

You will have a feeling of how the transition state looks like structurally, yeah.

Yeah.

So, those are the next steps for our project, in addition to what you said, which was the contact map analysis, and using cut-offs for the contact map, as well as changing the MJ matrix.

For...

And then I had one last thing to discuss with you. Can I share my screen? Okay. Yeah, yeah, sure.

Yeah, I think you can share the screen now. So, I sent a link for a competition that I want to apply for. I had mentioned it in a previous meeting, but I think you didn't know because it's a high school award.

And one thing I wanted to discuss is I think that our project is mainly physics-based, because it's simulations, and it uses, like, thermodynamics and, like, attractions, but,

It says biology includes mathematical modeling of biological systems, so would you say that our project is physics, or a mathematical modeling of biological systems?

It's actually, hard to see, but, yeah, yeah, but on, just from this page,

I would say that...

our project is more, belonging to the biology, because you have... it explicitly incites the mathematical modeling of biological system. But what we are doing is actually, the biological physics is... because this... this...

If... if you see the two frames, one is biological physics, the other one is physical biology.

You may think they are very similar, but it's totally different. One is emphasizing the physics, the other one is emphasizing the biology. So our project is actually, as you mentioned, I totally agree with you, it's actually emphasizing the physics rather than biology. But on this page.

Since... they already explicitly point, point out the biology includes the mathematical modeling of the biological system.

I think, perhaps, it would be better to,

Does this have other category, like, multi-interdisciplinary, no, these are the subject categories, like, on the page, it's only these...

6. Let me... let me think about that, okay,

And when you apply this award, you are going to present, or you just write down the application material? No, we have to submit our own projects, so first I have to submit an abstract for our project.

And then, after that, I can... I have to submit the research paper.

Of what we have. The search paper. Yeah.

So, after, after being... being awarded, you have to submit, or before?

I have to submit the abstract before, and then the research paper later.

to research paper to, to the, like, this, this, I, I don't know, to, to, to the journal, or to.

To this organization. So, you apply it to this competition.

So you submit your research paper to this competition? This isn't a journal, it's... and then they will review and cross-examine your research paper. Okay, okay, I understand. Let me think about that, including biology, mathematical modeling, biological system.

Engineering Physics.

So what do you prefer?

I think because they already said mathematical modeling of biological systems, I think that would be... that it means that we have to apply in biology, because they haven't said anything for physics.

Yeah.

But one thing is, I think they have a, like, a selection committee, committee in this, in this, computation. And this committee, I think, I support they are, this committee members, I support they are coming from different,

Research field. Yeah. And,

If... if your application material, is given to,

biological background researcher. I... I'm a little worried that maybe, your work And...

Cannot be well appreciated.

That's, that's, that's, my, my feeling, because, for biology people, they always care about this, this application.

Right? So you observe something, so how... they will... they will simply ask you the situation that, how realistic is your model?

And what kind of application you can...

You can give to the, the, the life science.

So this is the... this actually happened to me. When I'm trying to apply for some, like, grant. And I always chose physics, not biology, because I would... for several times, I wouldn't say a couple of times, but several times, I had submitted whatever grant application or some other application to the biology people, and they always ask a very...

very, specific biological problem. Like, you know, your model is good, but, how the biology can benefit from the model? And, this is something,

I'm a little worried about that. So I'm not very sure, about the background of this committee member, but, I just, hope my situation

My previous, appearance will not happen for you. Maybe I'll plan to ask them and give them what my project is a little bit about, and they can tell me if

I can apply in physics, maybe.

Yeah, yeah, yeah. I think you can... and in that situation, you can have more freedom to write down your abstract, your research paper, because I will assume that you will spend a

lot of, species on words for, like, introducing the background of the project, which is more physics rather than biology, I think.

Yeah, yeah. Okay, so, talking about the research paper, when can we, start writing the research paper or concluding our work after the next steps that we have discussed today?

I think, one direction, you proposed is very good for the PE fold, but make sure that, there was not showing in the previous study, so you have to...

search online, and you can also talk to the chat GPT.

And I think ChatGB can also, help you to find whether this, kind of research has been done before. This is the one direction. The second direction is, is the randomly distributing, the MGM metric. And after that, I think we are done, and we can actually, but, but.

After that, you can also do this P-fold analysis for that, you know, randomly... Yeah, yeah, yeah, yeah, yeah. And after that, we are done, actually. I think the whole story is there, and what we want to see that this non-native interaction can benefit... can facilitate the protein fold, and, how...

how the nature, manages this is in the sequence, right? When you try to change the sequence information, and the protein can still fold, with the native topology, but it cannot fold in so efficient. So the nature, actually optimized the...

the native interaction and the non-native interaction simultaneously. That's our, proposal.

And let's see if we can find the result consistent with our proposal. If that are consistent.

I think we are done. It's not a, it's not sophistic, so,

high-level, I would say, because I've been working in this, in this field for more than, I think, 15 years, I wouldn't say that it's a high-level study, but it's a very interesting study. I, I think, in particular for, for, for you, and, I, I mean...

you are not doing your PhD degree, you are not doing your PhD study and research, I think it would be a very good study. A very interesting study.

Okay? So is this something that you will also co-author?

Yeah, I would like to... I think, after that, we can submit together to the journal, right? And I think that would be better, also better for you, because if we, submit together, and I would assume that we have a higher, a little bit higher chance to be accepted.

Because I have been working in this field for many years, and I know how to deal with this submission, and how to find a good journal to submit.

But one thing I would like to, mention is that, finally, if we have all the results, we have to think about how to organize it, you know? Currently, we have a good result, but the picture does not look so...

good. You know that when you look at, the, the published paper, in particular in the good journal, you will see that their plot, looks, their plot looks so beautiful, right?

Yeah. So we have to think about a way to improve the, the, the, the figure. So how do you typically, how do you, when you publish papers, what do you use to make it look better? I... I use a lot of, software. One is called Orin. I'm not sure if you know that. The other one, I think, you know how to write the Python, right? Yeah. The Python, and there is a software called plot. Yeah, plot web, yeah. I also use GNUplot,

But Zalo is, like, an old days, old days software, but currently, my, my student, my PhD student, is mostly using, our, the, the multiplot.

I think Metiplot should be enough. And I also use MetaLab. But Metalab, I think all the function, in the MetaLab has been, implemented in the metaphlot.

So you can use that one. And, and in my, according to my, experience, it's,

Currently, we don't have to worry about the figure presentation.

We just have the result, and then we think about how to organize the story. And, after, we know, the, the, the, the story of our, of our project, and then we will try to improve the figure presentation. So that's the... I think that will be the workflow, okay?

Yeah. Okay, it looks very good. Hopefully, we can, we can make everything done in half of a year, with just, you know, something. That would be very good, I think.

Okay. I'm actually proud of you, and I think you have done a lot of good work, and you also, have a very,

You also think very hardly, hard to find out the very good direction to go. Honestly speaking, I didn't think about the P4, but you have

gave me a good reminder that we can actually do the pre-fold analysis. Just make sure that no one have done it before, otherwise...

you know, if this had been done before, there's no reason for us to do it again, right? Okay.

Okay, good. Okay. Yeah, anything else?

Boom.

No, not that I can think of. I'll message you when I need anything. Yeah, and, and, regarding this application, this is a work application, if you want to have, any help or assistance from myself, you just, you know, give, let me know, like, you know, if you, write down the abstract and,

That you can send to me. I'm not sure if it's okay. According to this, you have to do it, by your... all by yourself, or you can? No, I think you can have... you can work with a mentor or, like, someone from a university. Oh, okay. Okay, okay, so this is okay for, for me to read and, give you a suggestion, right? Yes. Okay. Okay, good, good.

Okay. Alright. Okay.

So let's, let's, keep in, touch, frequently, and whenever you have, and a progress is just, put in the, in the WeChat, or send me, send to me the email. The summary of this meeting, of, like, what the next steps are, I'll put them all together.

Yeah, yeah, good, good. But, one thing I have to apologize, because, I'm very busy, in particular, in the next semester, I'll be, having a very, heavy, teaching duty, so I may not respond to you so efficiently. In some cases, you just,

you just send me an email. If I don't... if I, do not repeat

if I do not respond to you in the WeChat, you just send to me the email, because some days I will, you know, check my email again, and finally I

I realize that someone has sent me the email. Because in WeChat, every day I receive so many messages, you know, I cannot, in particular for your kids, when you, when you send me a lot of results, a very good result, I have to, I have to spend time to read, but, in some situation, at that moment, I may not have time to read carefully, so I just missed this informed message. So you can, so my suggestion is, perhaps, if you have a very fast question.

That you can send to... you can, you can... you can put your, your, your question, on the WeChat. But if you organize all the things, like, like the one you did before, like the, you know, electronic maps, and a lot of graphs, you can send to, me the email. I think that would be better.

If there are any issues, like, just, like, in running some sort of simulation, or, like, coding-wise, can I just contact the PhD student?

Yeah, you can just put it in the, in the WeChat group. I talked to him, and, he would be very glad to help you. But, I think...

Maybe he, he should also be the co-author if, he, spend, you know, a lot of time and energy, in that, in your project, right? And I can also help you to, revise the manuscript, I think it would be good, and,

Yeah, yeah, you don't have to, contact private, him, privately, you just put all this, information in the, in the chat, because I, I, I, I also want to know,

How much effort he...

have put in your project, right? And that's... I cannot see that probably, you know, he helped you a lot, and finally, I do not know, and I cannot, put him as the co-author of the... of your research paper, right? You just put it,

the, your, your question and, your, your, your, any, your short simulation results on this WeChat, WeChat, but, but when you, try to organize a long summary or something, you just send me an email. Email, yeah. Okay? Okay.

Okay, good.

Okay, and else? I didn't say anything else?

Okay, okay, okay, let's, let's keep you in touch, and I will see you sometime. Thank you.

Okay, bye-bye. Bye.